EuqVid-Text

A Dialogue-Based Approach to Gender Bias Detection in Video Datasets

Component C3 (Textual) · EuqVid: A Multimodal Approach to Gender Bias Detection in Video Datasets
Group J26-DS-343 · CoEAI · SLIIT · IT4010 Research Project
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> **Status**
> *Draft for group discussion and supervisor review. Nothing in this document is final. Section 10 lists the decisions that must be settled before extraction begins, and Section 9 lists the risks that could invalidate the component if they are not addressed.*

1. Purpose of this document

This specification answers four questions about the textual component of EuqVid: what text means in the context of a video dataset, why text is a legitimate carrier of gender bias, what prior research establishes that claim, and how the component turns raw text into a bias score. It is written so that a reader who has never seen the project can follow the argument from premise to output.

2. What “text in a video” means

Video datasets do not only ship pictures. They ship words alongside the pictures, and those words come in four distinct kinds. Only two of them are relevant to this component, and confusing them is the fastest way to build a metric that measures nothing in particular.

Kind of text	What it is	Where it comes from	Available in our corpora
Dialogue	The words characters actually speak to each other.	Transcripts shipped with the dataset, subtitles, screenplays, or ASR output.	MELD (utterance CSVs with speaker labels); AMI (manual transcripts).
Descriptions	What a narrator or annotator states is happening on screen. Written for blind and partially sighted viewers.	Audio-description tracks; video captioning annotations.	LSMDC (with per-character gender labels; access application pending).
On-screen text	Signs, titles, chyrons and burned-in captions visible inside the frame.	Optical character recognition on the frames.	Not relevant to narrative gender representation. Excluded.
Metadata text	Character names, plot summaries, genre tags, cast lists.	Dataset annotation files.	Supporting role only; used for the speaker-to-gender mapping.

So **text in a video means the words attached to the video, not letters visible inside the frame**. The two candidates for this component are dialogue and descriptions. They are not interchangeable: dialogue measures how a character *speaks*, descriptions measure how a character is *written about*. Mixing them produces a feature that measures neither. Section 10 records this as the first decision to settle.

3. Why text carries gender bias

The clearest way to see what text contributes is to hold a single scene and read all four modalities off it at once.

> **One scene, two characters**
>
> ***Visual*** *— his face is large and centred; hers is small and at the edge of frame.*
> ***Temporal*** *— he speaks for 40 seconds, she speaks for 8.*
> ***Audio*** *— his voice sits above the music; hers sits underneath it.*
> ***Text*** *— he says "Get the car, we're going after him." She says "Please be careful."*

The first three modalities measure **how much room** each character is given. Text measures **what kind of person the script made them**. That distinction is what makes the component non-redundant: equalise her screen time, her speaking seconds and her loudness, and she is still the character who says *please be careful*. No visual, temporal or acoustic feature can detect that difference, because the difference is not in how much she is present but in what she is given to do.

Textual gender bias operates at four levels, in increasing order of depth.

Level	What is measured	Example of the disparity	Difficulty
1. Volume	How much each gender speaks.	Male characters are given the majority of dialogue in most films.	Low - word counting.
2. Interaction	Who speaks to whom.	Female characters rarely converse with each other; the Bechdel test formalises this.	Medium - needs addressee inference.
3. Subject matter	What each gender talks about.	Female dialogue concentrates on relationships and family; male dialogue on work and action.	Medium - topic modelling or lexicons.
4. Agency and power	Whether a character acts or is acted upon, and whether their language positions them above or below others.	Female characters are written with systematically lower agency and lower power.	High - needs dependency parsing plus a connotation lexicon.

Level 4 is where the component earns its novelty. Levels 1 to 3 are countable and largely confirm what the visual and temporal components already suggest. Level 4 measures something no other modality can reach.

3.1 Why text bias in a video dataset matters for model training

An examiner will reasonably ask why text is treated as a property of a *video* dataset. The answer is that video datasets are consumed multimodally. MELD exists precisely to train models that read text, audio and video together; video-language models, captioning systems and video question-answering systems all ingest the transcript alongside the frames. Bias present in the text half of a training corpus therefore propagates into the trained model exactly as bias in the visual half does. Section 4.2 gives the published evidence for that propagation, including the finding that models do not merely inherit dataset bias but amplify it.

4. Evidence from prior research

> **Verification required before citation**
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> *Every reference in this document was recalled from the literature, not retrieved from a database in this session. Each entry carries a confidence flag.* ***Verify author list, title, venue and year on Google Scholar or the ACL Anthology before any of these enter the TAF or the thesis.*** *Do not cite a specific numerical finding until you have read the paper and confirmed the number yourself.*

4.1 Established: gender bias is measurable in film and television text

Study	Venue / year	What it establishes	Conf.
Sap, Prasettio, Holtzman, Rashkin & Choi — *Connotation Frames of Power and Agency in Modern Films*	EMNLP 2017	Analysed a large set of modern film scripts and found female characters written with systematically less power and less agency than male characters. The connotation-frame lexicons were released publicly. **This is the anchor citation for the component and the source of the Level 4 features.**	High
Agarwal et al. — automating the Bechdel test	NAACL 2015	Built a computational version of the Bechdel test over film scripts, establishing that interaction-level gender structure is machine-detectable.	Med-High
Ramakrishna, Martínez, Malandrakis, Singla & Narayanan — linguistic analysis of movie character portrayal	ACL 2017	Quantified linguistic differences in how male and female movie characters are written.	Med-High
Bamman, O'Connor & Smith — *Learning Latent Personas of Film Characters*	ACL 2013	Learned character archetypes from film text; the recovered persona clusters are strongly gendered.	High
Madaan et al. — gender stereotyping in Bollywood films	FAT* 2018	Detected and quantified gender stereotyping from film text and images, demonstrating the method transfers outside Hollywood.	Medium
Fast, Vachovsky & Bernstein — *Shirtless and Dangerous*	ICWSM 2016	Quantified linguistic signals of gender stereotyping in narrative fiction writing.	Medium
Anderson & Daniels — film dialogue by gender across roughly 2,000 screenplays	The Pudding, 2016	Men hold the majority of dialogue in the large majority of films. **Data journalism, not peer reviewed — use for motivation only, never as an empirical anchor.**	High*

4.2 Established: text bias in training data transfers into models, and amplifies

This is the more important half of the evidence base. It is what justifies measuring bias *before* training rather than repairing the model afterwards, and it supports the whole of EuqVid, not only this component.

Study	Venue / year	What it establishes	Conf.
Zhao, Wang, Yatskar, Ordonez & Chang — *Men Also Like Shopping*	EMNLP 2017	Models trained on gender-skewed data are **more biased than the data itself**. Bias amplification is the single strongest argument for a pre-training dataset metric: a corpus that is moderately skewed yields a model that behaves as though it were severely skewed.	High
Hendricks, Burns, Saenko, Darrell & Rohrbach — *Women Also Snowboard*	ECCV 2018	Captioning models predicted a subject's gender from scene context rather than from the person, reproducing dataset stereotypes at inference time. Direct evidence that gendered text in training data becomes gendered model behaviour.	High
Caliskan, Bryson & Narayanan	Science, 2017	Human-like biases are demonstrably embedded in language corpora themselves, independent of any particular model.	High
Bolukbasi, Chang, Zou, Saligrama & Kalai — debiasing word embeddings	NeurIPS 2016	Gender stereotypes are encoded in representations learned from text corpora.	High

4.3 What the literature does not yet provide

Lining the three findings up exposes the gap precisely:

- **Known.** Film and television dialogue is gendered in volume, in interaction structure, in subject matter and in agency. Multiple peer-reviewed studies, listed in 4.1.
- **Known.** Models trained on gendered text inherit that bias and amplify it. Listed in 4.2.
- **Missing.** No method turns these findings into a number a dataset builder can compute on a *video* dataset, before training, on the same scale and in the same framework as the visual, temporal and acoustic signals.

The natural language processing community has measured film text bias. The computer vision community has measured screen-time bias. Neither has combined them into a single pre-training dataset audit score. That is the space this component occupies.

5. Problem, gap and novelty

	Statement
Problem	Video datasets used to train multimodal models carry gender bias in their text track as well as their visual track. That textual bias is invisible to existing dataset documentation, and because models amplify the bias they are trained on, it surfaces only after training as measurable performance and representation disparity, when correction is expensive.
Gap	Textual gender bias in film and television has been studied extensively in NLP, but always as a descriptive finding about a corpus of scripts. It has never been formulated as a computable, scale-invariant dataset-level score, and never integrated with visual and temporal signals into a single pre-training audit metric for video datasets.
Novelty	The first dialogue-level gender disparity score for video datasets, combining volume, interaction, subject-matter and agency-power signals into a frozen closed-form equation on a shared -10 to +10 scale, designed to compose with the visual, temporal and acoustic sub-metrics of EuqVid rather than standing alone.
Scope limit	Narrative multi-person video: film and scripted television, in English, where two or more characters exchange dialogue. The component measures *within-content* disparity: given characters of both genders in the same material, is one systematically given more to say and more agency in saying it? Zero denotes parity, not the naturally occurring distribution.

6. Extracted features

All features are computed per gender per shard and expressed as **shares or ratios, never raw counts**. This is the same scale-invariance requirement that governs every other EuqVid component: a raw word count grows with corpus size, so an equation fitted on raw counts would score a large balanced dataset as more biased than a small skewed one.

> **Means, not standard deviations**
>
> *EuqAud used standard deviations of pitch, energy and amplitude because those quantities are intrinsically gender-linked — a mean would merely detect that a woman is present. Dialogue volume and agency are not intrinsically gender-linked, so that rationale does not transfer. The bias signal here lives in the mean, and taking a standard deviation would remove it. This component uses means. The same argument applies to the visual component's spatial features.*

#	Feature	Definition	What it captures	Lvl
F1	dialogue_volume_share	Words spoken by gender g divided by total words in the shard.	Raw speaking allocation in language terms.	1
F2	utterance_count_share	Utterances by gender g divided by total utterances.	How often a gender is given the floor, independent of length.	1
F3	mean_utterance_length	Mean words per utterance for gender g, normalised by the shard mean.	Whether one gender gets substantive lines and the other gets short reactions.	1

#	Feature	Definition	What it captures	Lvl
F4	agency_score_mean	Mean agency value of verbs for which a character of gender g is the syntactic subject, looked up in the connotation-frame lexicon.	Whether characters act or are acted upon.	4
F5	power_score_mean	Mean power value over the same subject-verb pairs.	Whether a character's language positions them above or below their interlocutor.	4
F6	same_gender_ dialogue_share	Proportion of gender g's utterances occurring in an adjacent turn with another character of the same gender.	Bechdel-style interaction structure. Approximate - see limitation L5.	2
F7	mention_ratio	Number of times a character of gender g is named by other characters, divided by the number of times they speak.	Whether a gender is talked about rather than talking.	2
F8	sentiment_valence_ mean	Mean sentiment polarity of gender g's utterances. Ships free with MELD as emotion and sentiment labels.	Emotional register assigned to each gender.	3

Eight raw features become six to seven disparity features after screening for collinearity (F1 and F2 will correlate strongly and one will likely be dropped inside cross-validation folds). At polynomial degree 2 over six features the expansion carries 28 coefficients, which is comfortable against a target of 200 or more shards. This is a substantial improvement on the 91-coefficient problem the single-equation design faced, and is one of the strongest technical arguments for the four-component structure.

7. Component flow

Twelve steps, from raw corpus to a frozen equation. Each step names its input and output so the component can be handed to another member or picked up after a break without reconstruction.

Step	What happens	Input → Output
Step 1 Acquire time-aligned text	For each corpus, produce a table of (clip_id, utterance_id, speaker_id, text, start_time, end_time). MELD requires no video processing at all - the annotation CSVs carry Utterance, Speaker, StartTime, EndTime, Dialogue_ID and Utterance_ID directly. AMI requires parsing the manual annotation XML and joining words to speaker segments.	Raw corpus → aligned utterance table
Step 2 Assign gender to each speaker	Map speaker_id to gender in {male, female, unknown}. For MELD this is a manual mapping from character name to gender for the main and recurring cast, published as an appendix. Utterances by unknown or ambiguous speakers are excluded and the excluded share is reported, never silently dropped. **This mapping is a shared group asset.** All four components must aggregate over the same identity-to-gender map, or the four sub-scores describe different populations and cannot be fused.	Utterance table → gendered utterance table
Step 3 Assign utterances to shards	Apply the group shard contract: target 200 or more shards, at least 30 clips per gender per shard, partitioned by identity so that no character spans two shards. See risk R2 - this contract is in direct tension with MELD's structure and must be resolved before extraction.	Gendered utterances → shard-assigned utterances
Step 4 Preprocess and parse	Tokenise, sentence-split and dependency-parse with spaCy. Strip non-speech markers and annotation artefacts. Retain the parse trees - Step 5 needs subject-verb relations, not bag of words.	Text → parsed text

Step	What happens	Input → Output
Step 5 Compute per-utterance features	Word counts for F1 to F3. For F4 and F5, extract verbs whose syntactic subject is the speaking character and look each verb up in the connotation-frame lexicon to obtain agency and power values; record the out-of-vocabulary rate. For F6, use turn adjacency. For F7, string-match known character names inside utterance text. For F8, read MELD's shipped sentiment labels.	Parsed text → per-utterance feature rows
Step 6 Aggregate to shard by gender	Sum or average each feature within each (shard, gender) pair.	Utterance rows → shard-gender feature table
Step 7 Convert to scale-free disparity features	Express every feature as a share or a normalised ratio between the male and female aggregates. No raw magnitude enters the model.	Shard-gender table → shard-level disparity vectors
Step 8 Generate ground-truth bias labels	Apply the **shared group augmentation protocol** - not a component-private one. Controlled incremental reduction produces shards at known disparity levels. If each component invents its own label definition, the four sub-metrics are calibrated against four different meanings of 'bias level' and the fused score is incoherent.	Disparity vectors → labelled training set
Step 9 Fit the sub-metric	Polynomial Ridge regression, degree capped at 2. Feature screening performed *inside* cross-validation folds, never on the full set beforehand. Grid search over the Ridge alpha. Report both nominal row count and effective sample size, the latter being the number of independent shards.	Labelled set → fitted model
Step 10 Freeze, normalise and tier	Algebraically fold the scalers back so the equation prints in raw feature units. Write the min-max transform to -10 to +10 as an explicit closed form with stated bounds. Derive the Neutral, Moderate and Strong thresholds from stated disparity levels rather than inheriting EuqAud's 2 and 6.	Fitted model → frozen closed-form equation
Step 11 Characterise stability and generalisation	1,000-iteration bootstrap resampled at shard and identity level, never at row level, which would produce fictitiously narrow confidence intervals on synthetically expanded data. Report score-level and tier-level stability. Add leave-one-corpus-out cross-validation and a monotonicity check along held-out perturbation curves.	Frozen equation → stability and generalisation evidence
Step 12 Hand off	Deliver the frozen equation, the normalisation constants, the tier thresholds and the per-shard scores to the fusion stage and to the validation component.	Equation → Text-EuqVid sub-score column

8. Final outcome

The component delivers six artefacts. Each is assessable on its own, without the fusion stage existing and without any other component being finished.

	Deliverable	Form
D1	Frozen closed-form Text-EuqVid equation in raw feature units, with all scaling constants folded into the published coefficients.	Equation plus coefficient table
D2	Per-shard textual bias scores on the -10 to +10 scale with Neutral, Moderate and Strong tier assignments, and the derivation of the tier thresholds.	Scored CSV plus threshold justification
D3	Ranked feature table reporting which textual signals carry bias information and which do not, with Pearson and Spearman correlations against the label.	Results table
D4	Bootstrap stability table: mean, standard deviation and 95 per cent confidence interval at score and tier level, with the resampling unit stated.	Results table
D5	The speaker-to-gender mapping, the excluded-speaker share, and the connotation-lexicon out-of-vocabulary rate.	Appendix, shared with the group

	Deliverable	Form
D6	Limitations register: what the component cannot measure, and on which corpora it cannot run at all.	Written section

The headline claim the component can support, if the results hold: **textual gender disparity in a video dataset is measurable before training, on the same scale as visual and temporal disparity, using a frozen equation that requires no retraining and can be audited line by line.**

9. Assumptions, limitations and risks

These are stated here so they appear in the thesis as declared limitations rather than as discoveries made by an examiner.

	Risk or limitation	Consequence and response
R1	**No transcripts on AVA.** The film corpus in our set ships face boxes and speaking labels but no dialogue.	**Most serious risk in the component.** The text sub-score cannot be computed on film, which is where the bias is expected to appear. Obtaining dialogue via ASR plus diarisation on mixed movie audio would introduce an error rate large enough to swamp the signal. Requires the decision in Section 10.1.
R2	**The shard contract conflicts with MELD.** Identity-level partitioning requires that no character spans two shards, but Friends has six leads who appear across almost every episode.	Two hundred identity-disjoint shards cannot be built from six recurring identities. Either shard by episode or scene and document the identity leakage honestly, or accept far fewer shards and revisit the coefficient budget. This affects all four components, not only this one.
R3	**MELD is a negative control.** Main-cast speaking-time ratio is 1.094 male to female; Friends is an unusually balanced ensemble.	Expect a near-zero score. This is the intended result and should be framed in advance as a false-positive test - a metric that correctly scores a balanced corpus as Neutral is evidence of specificity - not presented afterwards as a disappointing finding.
R4	**Lexicon domain mismatch.** The connotation-frame lexicon was built on film screenplays; MELD is informal, overlapping television conversation.	Coverage gaps are expected. Measure and report the out-of-vocabulary rate for subject-verb pairs. If coverage is poor, F4 and F5 weaken and the component falls back to Levels 1 to 3.
R5	**Addressee is approximated.** F6 infers who is being spoken to from turn adjacency, not from actual addressee annotation.	In multi-party scenes with three or more speakers this approximation degrades. State it as an approximation and consider restricting F6 to two-speaker exchanges.
R6	**Gender is derived, not given.** On MELD, gender comes from a manual character-name mapping.	The component inherits any error in that mapping. Non-binary and ambiguous characters are excluded rather than forced into a binary, and the excluded share is reported. The binary framing is itself a declared limitation of the whole project.
R7	English-language corpora only.	Findings do not transfer to other languages without re-validation. State in scope.
R8	**Construct ambiguity between dialogue and descriptions.**	How a character speaks and how a character is described are different variables. They cannot be averaged into one feature. Section 10.1 must resolve which one the component measures.

10. Open decisions to settle before extraction

	Decision	Why it must come first
10.1	**Dialogue or descriptions?** Dialogue means MELD and AMI. Descriptions means LSMDC, which is the only corpus in the set with explicit per-character male/female labels but whose access application is still pending.	It determines which corpora the component runs on, whether the fused EuqVid score can be computed on film at all, and whether the component title is accurate. Everything else follows from it.

	Decision	Why it must come first
10.2	Does the shard contract survive MELD's six-identity structure, and what is the agreed fallback?	Determines the number of independent rows, which determines whether the regression is fittable. Cost of agreeing now is a naming convention; cost of discovering it later is full re-extraction by four people.
10.3	Is the augmentation and labelling protocol shared across all four components?	Four private label definitions cannot be fused into one score. This must be a group deliverable.
10.4	Are the four sub-scores calibrated on one shared reference set?	Four independently min-max normalised scores are not comparable: +6 from one component does not denote the same disparity as +6 from another.
10.5	Who owns the fusion stage and the cross-modal validation?	Under the four-modality structure this work currently has no named owner. Unowned work is typically unassessed work.

11. References

*Recalled from the literature and **not yet verified against a database**. Confirm every entry before citation. Ordered by relevance to this component.*

	Reference	Conf.
[1]	Sap, M., Prasettio, M. C., Holtzman, A., Rashkin, H., & Choi, Y. (2017). Connotation Frames of Power and Agency in Modern Films. *Proceedings of EMNLP 2017*.	High
[2]	Zhao, J., Wang, T., Yatskar, M., Ordonez, V., & Chang, K.-W. (2017). Men Also Like Shopping: Reducing Gender Bias Amplification using Corpus-level Constraints. *Proceedings of EMNLP 2017*.	High
[3]	Hendricks, L. A., Burns, K., Saenko, K., Darrell, T., & Rohrbach, A. (2018). Women Also Snowboard: Overcoming Bias in Captioning Models. *Proceedings of ECCV 2018*.	High
[4]	Caliskan, A., Bryson, J. J., & Narayanan, A. (2017). Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334).	High
[5]	Bamman, D., O'Connor, B., & Smith, N. A. (2013). Learning Latent Personas of Film Characters. *Proceedings of ACL 2013*.	High
[6]	Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V., & Kalai, A. (2016). Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings. *Advances in Neural Information Processing Systems*.	High
[7]	Agarwal, A., et al. (2015). Key Female Characters in Film Have More to Talk About Besides Men: Automating the Bechdel Test. *Proceedings of NAACL-HLT 2015*.	Med-High
[8]	Ramakrishna, A., Martínez, V. R., Malandrakis, N., Singla, K., & Narayanan, S. (2017). Linguistic analysis of differences in portrayal of movie characters. *Proceedings of ACL 2017*.	Med-High
[9]	Park, J. S., Darrell, T., & Rohrbach, A. (2020). Identity-Aware Multi-Sentence Video Description. *Proceedings of ECCV 2020*. arXiv:2008.09791. *Source of the LSMDC per-character gender annotations.*	Med-High
[10]	Madaan, N., et al. (2018). Analyze, Detect and Remove Gender Stereotyping from Bollywood Movies. *Proceedings of FAT* 2018*, PMLR 81.	Medium
[11]	Fast, E., Vachovsky, T., & Bernstein, M. S. (2016). Shirtless and Dangerous: Quantifying Linguistic Signals of Gender Bias in an Online Fiction Writing Community. *Proceedings of ICWSM 2016*.	Medium
[12]	Gala, D., Khursheed, M. O., Lerner, H., O'Connor, B., & Iyyer, M. (2020). Analyzing Gender Bias within Narrative Tropes. *NLP+CSS Workshop, EMNLP 2020*.	Medium
[13]	Anderson, H., & Daniels, M. (2016). Film Dialogue from 2,000 Screenplays, Broken Down by Gender and Age. *The Pudding*. **Data journalism, not peer reviewed.**	High*
[14]	Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., & Mihalcea, R. (2019). MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations. *Proceedings of ACL 2019*.	High

	Reference	Conf.
[15]	Geena Davis Inclusion Quotient (GD-IQ). Google, USC Viterbi School of Engineering, and the Geena Davis Institute on Gender in Media. *Screen-time and speaking-time disparity in top-grossing films.*	High
[16]	Smith, S. L., et al. USC Annenberg Inclusion Initiative annual reports on inequality in popular films. *Confirm the specific year and report edition before citing a figure.*	High
[17]	Jayawardena, S., et al. (2026). EuqAud. *IEEE Access*. *Parent methodology. Section V names multi-speaker audio, not video, as future work; Section VI says multimodal data.*	Verified in project

Prepared as a working specification for Group J26-DS-343. All empirical claims attributed to prior work require verification against the source before they are repeated in the TAF or the thesis. No results in this document are our own measurements; the only measured figure referenced is the MELD main-cast speaking-time ratio of 1.094 male to female, computed by the group.